

Relative Cowedges and Coends over a Site

Relative (co-)Yoneda Lemma for Categories Indexed over a Small-Generated Site

Indexed categories, fibrations, stacks and relative category theory

Indexed categories for relative topos theory

Yiqi

August 25, 2026

Abstract

This preview contains the foundational and definitional stages of the relative-coend programme. It fixes the base, variance, coherence and source boundaries, defines localized relative cowedges and coends, and proves their first formal properties. It is a stage preview, not a release of the complete article.

Contents

1	Introduction	2
2	The ambient indexed data	3
2.1	Sites and relative-topos conventions	3
2.2	Indexed categories, opposites, and coefficients	3
3	Localized cowedges and relative coends	4
4	Restriction and stable base change	5
5	First comparison theorems	6
6	Interface with relative topos theory and prior calculus	7
6.1	The canonical stack as coefficients	7
6.2	Relation to Shulman’s indexed profunctors	8
6.3	Presentation boundary	8
7	Next stages and precise proof obligations	8

1 Introduction

Ordinary coends package dinatural families into a single universal object. They support the calculus of Kan extensions, profunctors, density, and the co-Yoneda lemma. For a category varying over a base, however, there are two different operations that one might call a coend. One can form an ordinary coend separately in each fibre, or one can ask for a universal object that is compatible with every change of base. The first operation is familiar but does not, by itself, produce the second. The elementary example in [example 5.3](#) makes this distinction explicit.

This paper begins a coend calculus adapted to categories indexed over a Grothendieck site. Its basic move is to formulate the universal property over the whole slice site $\mathcal{C}/U$. A relative cowedge with apex X therefore has a component not only for every object in $\mathbb{D}(U)$, but for every generalized object $A \in \mathbb{D}(V)$ above every arrow $v : V \rightarrow U$. A second axiom compares these components after further pullback. The resulting universal object records precisely the descent-sensitive data that a fibrewise definition omits.

The motivation comes from Drioux’s lecture series on indexed categories and relative topos theory [[Dri25b](#), [Dri25a](#), [Dri26a](#), [Dri26b](#)]. Those slides give an indexed Yoneda construction, formulate indexed colimits by localization to every slice, and announce relative (co-)Yoneda and indexed-(co)limit projects. They are programme-level sources, not published proofs of the claims developed here. We therefore reconstruct the definitions from their universal properties and mark every later existence or computation statement as an open problem.

There is important prior art. Shulman’s indexed-enriched formalism already constructs a canceling tensor product and the composition of indexed profunctors by a fibrewise coequalizer under indexed coproduct and Beck–Chevalley hypotheses [[Shu13](#), §§2–3]. Lee develops indexed profunctors over 2-categories [[Lee23](#)]. The present definition is narrower in one direction and different in another: it is initially unenriched and tied to a fixed site, but it makes localization over every slice part of the cowedge itself. Thus the intended new problem is not merely to decorate the ordinary coend sign. It is to relate coend calculus to topology, descent, and change of site in relative topos theory.

Results of phases 0–1

After fixing size and variance conventions, we:

1. define localized relative cowedges, coends, wedges, and ends;
2. identify a relative coend with a representing object for the cowedge functor;
3. construct restriction along every map of the site and the canonical base-change comparison between local coends;
4. prove invariance under a chosen indexed equivalence and reduction to the ordinary definition over the terminal site; and
5. exhibit a two-object indexed category for which fibrewise coends fail to assemble, while the localized coends are stable.

All these statements are proved in this paper. We do *not* yet claim a general existence formula, Fubini theorem, relative co-Yoneda lemma, profunctor bicategory, or invariance under a change of site presentation. Those are separated into explicit problems in [section 7](#).

2 The ambient indexed data

2.1 Sites and relative-topos conventions

Fix a Grothendieck universe and a small site $(\mathcal{C}, J)$. Write

$$\mathcal{E} = \mathbf{Sh}(\mathcal{C}, J), \quad \ell_J = a_{JY} : \mathcal{C} \longrightarrow \mathcal{E}$$

for the associated sheaf topos and the sheafified Yoneda functor. The canonical stack attached to this presentation is the pseudofunctor

$$\mathbb{S}_{(\mathcal{C}, J)} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{CAT}, \quad U \longmapsto \mathcal{E}/\ell_J(U),$$

whose reindexing functors are pullbacks. Caramello and Zanfa prove that this is a J -stack [CZ21, Definition 2.6.4 and Theorem 2.6.8]. It will be the principal coefficient system for applications, although the definition below works for a general indexed category of coefficients.

In the terminology of relative topos theory, a relative topos over $\mathcal{E}$ is a geometric morphism $p : \mathcal{F} \rightarrow \mathcal{E}$ [CZ21, Definition 8.2.2]. We keep this distinct from an indexed category $\mathbb{D}$: the latter is input from which relative categorical constructions are formed, not by itself a relative topos.

The smallness assumption is a phase-1 convention. A small-generated site is locally small and has a small J -dense subcategory [CZ21, p. 4]. After choosing such a subcategory one may apply the same formulas to the resulting small presentation; independence of that choice is a separate comparison theorem and is not assumed below. This distinction is essential: a formula on a presentation is not automatically an invariant of the base topos.

Remark 2.1 (Where the topology enters). The equations in [definition 3.1](#) use the underlying category $\mathcal{C}$, not the covering families of J directly. The topology enters when $\mathbb{D}$ and $\mathbb{V}$ carry J -descent, when H respects that descent, or through the choice $\mathbb{V} = \mathbb{S}_{(\mathcal{C}, J)}$. Without such hypotheses the construction is localization-sensitive but not yet topology-invariant.

2.2 Indexed categories, opposites, and coefficients

Let

$$\mathbb{D} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{Cat}, \quad \mathbb{V} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{CAT}$$

be pseudofunctors. We assume that each $\mathbb{D}(U)$ is small and that each $\mathbb{V}(U)$ is locally small. For an arrow $u : U' \rightarrow U$, both reindexing functors are denoted u^* ; the ambient pseudofunctor will always be clear. The coherence isomorphism is written

$$\alpha_{v, u, X} : v^* u^* X \xrightarrow{\cong} (uv)^* X.$$

We suppress unit constraints when no ambiguity can result.

Let $\mathbb{D}^\vee$ denote the fibrewise opposite indexed category: its fibre at U is $\mathbb{D}(U)^{\text{op}}$, with the induced pseudofunctorial coherence. This is the indexed opposite used in relative category theory [CZ21, Definition 2.1.4]. Fix an indexed functor

$$H : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \longrightarrow \mathbb{V}.$$

Thus each fibre has a functor

$$H_U : \mathbb{D}(U)^{\text{op}} \times \mathbb{D}(U) \longrightarrow \mathbb{V}(U),$$

and every $u : U' \rightarrow U$ has an invertible comparison

$$\chi_{u, A, B}^H : u^* H_U(A, B) \xrightarrow{\cong} H_{U'}(u^* A, u^* B), \tag{1}$$

natural in A, B and coherent under composition. Requiring χ^H to be invertible is the point at which H is genuinely indexed rather than merely laxly indexed.

For $U \in \mathcal{C}$, the localization of $\mathbb{D}$ at U is

$$\mathbb{D}/U : (\mathcal{C}/U)^{\text{op}} \longrightarrow \mathbf{Cat}, \quad (v : V \rightarrow U) \longmapsto \mathbb{D}(V).$$

If $X \in \mathbb{V}(U)$, its cartesian constant section over $\mathcal{C}/U$ has value v^*X at $v : V \rightarrow U$. These two constructions supply the domain and apex of the relative cowedges below.

3 Localized cowedges and relative coends

Recall that for an ordinary functor $K : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \mathcal{B}$, a cowedge to X is a dinatural family $K(A, A) \rightarrow X$; its initial cowedge is the coend [Lor23, Definition 1.1.4 and Definition 1.1.6]. We add a base coherence condition to this familiar definition.

Definition 3.1 (Localized relative cowedge). Fix $U \in \mathcal{C}$ and $X \in \mathbb{V}(U)$. A *relative cowedge from H to X over U* is a family

$$\omega_{v,A} : H_V(A, A) \longrightarrow v^*X \quad (2)$$

for every arrow $v : V \rightarrow U$ and every $A \in \mathbb{D}(V)$, satisfying:

(C1) *fibrewise dinaturality*: for each $f : A \rightarrow B$ in $\mathbb{D}(V)$,

$$\omega_{v,A} H_V(f^{\text{op}}, 1_A) = \omega_{v,B} H_V(1_B, f) : H_V(B, A) \longrightarrow v^*X; \quad (3)$$

(C2) *base coherence*: for every $w : W \rightarrow V$,

$$\omega_{vw, w^*A} \chi_{w,A,A}^H = \alpha_{w,v,X} w^*(\omega_{v,A}) : w^*H_V(A, A) \longrightarrow (vw)^*X. \quad (4)$$

The set of such families is denoted $\text{Cow}_U^{\text{rel}}(H; X)$.

The definition ranges over the Grothendieck construction of $\mathbb{D}/U$, but writing out equations (3) and (4) avoids choosing a cleavage for that construction. The two equations have different jobs: dinaturality moves within one fibre, whereas base coherence moves between fibres.

A morphism $(X, \omega) \rightarrow (Y, \tau)$ is a morphism $m : X \rightarrow Y$ in $\mathbb{V}(U)$ such that

$$v^*m \omega_{v,A} = \tau_{v,A}$$

for all v, A . This defines the category $\mathbf{Cow}_U(H)$. Equivalently, $\mathbf{Cow}_U(H)$ is the category of elements of the covariant functor

$$\text{Cow}_U^{\text{rel}}(H; -) : \mathbb{V}(U) \longrightarrow \mathbf{Set}.$$

Definition 3.2 (Relative coend). A *relative coend of H over U* is an initial object (L_U, λ^U) of $\mathbf{Cow}_U(H)$. When it exists, we write

$$L_U = \int_{\text{rel}, U}^{A \in \mathbb{D}} H(A, A).$$

The notation records that the cowedge is localized over $\mathcal{C}/U$; it is not an assertion that an ordinary fibrewise coend has already been computed.

Proposition 3.3 (Representing property and uniqueness). *For $L_U \in \mathbb{V}(U)$, the following are equivalent.*

1. There is a relative cowedge λ^U making (L_U, λ^U) initial in $\mathbf{Cow}_U(H)$.
2. There is a natural isomorphism

$$\mathbb{V}(U)(L_U, X) \cong \mathbf{Cow}_U^{\text{rel}}(H; X) \quad (5)$$

sending 1_{L_U} to λ^U .

Consequently, a relative coend is unique up to a unique isomorphism compatible with its universal cowedge.

Proof. Given an initial cowedge, send $m : L_U \rightarrow X$ to the family $(v^* m \lambda_{v,A}^U)_{v,A}$. Initiality gives the inverse and its uniqueness, while composition in $\mathbb{V}(U)$ gives naturality in X . Conversely, the image of 1_{L_U} under [equation \(5\)](#) is a universal element, hence an initial cowedge. The final assertion is the standard uniqueness of representing objects, including compatibility with the distinguished universal element. $\square$

Definition 3.4 (Relative wedge and end). For an indexed functor $K : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \rightarrow \mathbb{V}$, a relative wedge from $X \in \mathbb{V}(U)$ to K consists of maps

$$v^* X \longrightarrow K_V(A, A)$$

that satisfy the equations dual to [equations \(3\)](#) and [\(4\)](#). A *relative end* is a terminal relative wedge. This draft takes the dual universal property as the definition; no completeness of the fibrewise opposite coefficient system is assumed.

4 Restriction and stable base change

The use of all generalized objects has an immediate payoff: cowedges restrict without any existence assumption.

Proposition 4.1 (Restriction to a slice). *Every arrow $u : U' \rightarrow U$ induces a functor*

$$\rho_u : \mathbf{Cow}_U(H) \longrightarrow \mathbf{Cow}_{U'}(H).$$

It sends an apex X to $u^ X$. Up to the pseudofunctorial constraints, the component of $\rho_u \omega$ at $v : V \rightarrow U'$ and $A \in \mathbb{D}(V)$ is $\omega_{uv,A}$. These functors are pseudofunctorial: $\rho_{1_U} \cong 1$ and $\rho_v \rho_u \cong \rho_{uv}$.*

Proof. Define

$$(\rho_u \omega)_{v,A} : H_V(A, A) \xrightarrow{\omega_{uv,A}} (uv)^* X \xrightarrow{\alpha_{v,u,X}^{-1}} v^* u^* X.$$

The dinaturality equation is exactly that for $\omega_{uv,-}$. Base coherence follows by pasting the base-coherence square for ω with the associativity coherence for α . On morphisms put $\rho_u(m) = u^* m$. The identity and composition comparisons are inherited from those of $\mathbb{V}$, and their coherence is inherited from the pentagon and triangle axioms of the pseudofunctor. $\square$

Suppose relative coends (L_U, λ^U) have been chosen for all $U \in \mathcal{C}$. The restricted cowedge $\rho_u \lambda^U$ has apex $u^* L_U$. Initiality at U' therefore supplies a unique comparison

$$b_u : L_{U'} \longrightarrow u^* L_U \quad (6)$$

such that $b_u \lambda^{U'} = \rho_u \lambda^U$, with the structural isomorphisms understood.

Definition 4.2 (Stable relative coends). A chosen family of relative coends is *stable under base change* if every canonical map b_u in [equation \(6\)](#) is an isomorphism.

Theorem 4.3 (Coherent assembly). *The canonical maps b_u satisfy the identity and composition laws up to the coherence of $\mathbb{V}$. If the relative coends are stable, the inverses*

$$b_u^{-1} : u^* L_U \xrightarrow{\cong} L_{U'}$$

make $(L_U)_{U \in \mathcal{C}}$ a cartesian pseudonatural object of $\mathbb{V}$. This structure is uniquely determined by the universal cowedges.

Proof. For an identity arrow, both the unit comparison and b_{1_U} are morphisms from the initial cowedge to its unit restriction; they agree by uniqueness. For composable arrows $U'' \xrightarrow{v} U' \xrightarrow{u} U$, the composite of b_v with $v^* b_u$, followed by the associativity constraint, and the map b_{uv} are morphisms from the same initial cowedge over U'' to the same twice-restricted cowedge. They again agree by uniqueness. If all b_u are invertible, reversing them gives the cartesian transition maps; the preceding identities are precisely their pseudonaturality axioms. Any other compatible transitions would invert morphisms with the same universal property and hence coincide with them. $\square$

Remark 4.4. Existence and stability are logically separate. The definition gives b_u only after relative coends exist on both slices; it does not force that map to be invertible. A useful existence theorem must therefore either prove stability or state it as an additional hypothesis.

5 First comparison theorems

Proposition 5.1 (Indexed-equivalence invariance). *Let $F : \mathbb{D} \simeq \mathbb{D}'$ be a chosen pseudonatural adjoint equivalence of indexed categories, including coherent unit and counit. Let $H' : (\mathbb{D}')^\vee \times_{\mathcal{C}} \mathbb{D}' \rightarrow \mathbb{V}$, and suppose an indexed natural isomorphism*

$$\theta : H \xrightarrow{\cong} H' (F^\vee \times F)$$

has been fixed. Then for every U there is an equivalence

$$\mathbf{COW}_U(H) \simeq \mathbf{COW}_U(H')$$

that is the identity on apex objects. In particular, one side has a relative coend if and only if the other does, and the resulting apex objects are canonically isomorphic relative to the chosen equivalence data.

Proof. Transport a component of an H' -cowedge along F_V and precompose with $\theta_{V,A,A}$. Naturality of θ gives fibrewise dinaturality, and its indexed coherence gives base coherence. Applying a chosen pseudoinverse of F gives a functor in the opposite direction. The unit and counit transport to natural isomorphisms between the composites and the identity; their triangle identities give an equivalence of cowedge categories. Because apexes and apex morphisms are unchanged, initial objects correspond and their apexes are uniquely isomorphic by [proposition 3.3](#). $\square$

The word *chosen* in [proposition 5.1](#) is deliberate. An unstructured statement that the fibres merely happen to be equivalent does not provide the coherence needed to transport [equation \(4\)](#).

Proposition 5.2 (Terminal-site reduction). *Let $\mathcal{C} = \mathbf{1}$ with its unique topology. If $\mathbb{D}(*) = \mathcal{A}$, $\mathbb{V}(*) = \mathcal{B}$, and $H_* : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \mathcal{B}$, then a relative cowedge over $*$ is exactly an ordinary cowedge for H_* . Consequently,*

$$\int_{\text{rel},*}^{A \in \mathbb{D}} H(A, A) \cong \int^{A \in \mathcal{A}} H_*(A, A)$$

whenever either side exists.

Proof. The slice $\mathbf{1}/*$ has one object and one arrow. Thus [equation \(2\)](#) is an ordinary family indexed by $A \in \mathcal{A}$, [equation \(3\)](#) is ordinary dinaturality, and [equation \(4\)](#) reduces to the unit axiom. The two cowedge categories are therefore isomorphic. $\square$

Example 5.3 (Fibrewise coends do not automatically assemble). Let $\mathcal{C}$ be the category $0 \xrightarrow{u} 1$, equipped with the trivial topology, and let $\mathbb{V}$ be the constant indexed category **Set**. Define

$$\mathbb{D}(1) = \emptyset, \quad \mathbb{D}(0) = \mathbf{1},$$

with the unique reindexing functor $u^* : \emptyset \rightarrow \mathbf{1}$. Let H_1 be the unique functor with empty domain and let $H_0(*, *) = \{*\}$.

The ordinary fibrewise coend is $\emptyset$ over 1 and a singleton over 0. Since u^* on **Set** is the identity, these two objects cannot be the values of a cartesian indexed object: there is not even a map $\{*\} \rightarrow \emptyset$ in the direction of [equation \(6\)](#).

The localized relative coends behave differently. Over 0, a relative cowedge to X is one map $\{*\} \rightarrow X$, hence a pointed set. Over 1, the fibre $\mathbb{D}(1)$ contributes no component, but the generalized object $* \in \mathbb{D}(0)$ above $u : 0 \rightarrow 1$ again contributes one point of X . Thus $\mathbf{Cow}_0(H)$ and $\mathbf{Cow}_1(H)$ are both the category of pointed sets. Their initial objects are singletons, and the base-change map is the identity. In this example the localization-sensitive definition restores stable assembly precisely by remembering the object visible after pullback.

6 Interface with relative topos theory and prior calculus

6.1 The canonical stack as coefficients

Taking $\mathbb{V} = \mathbb{S}_{(\mathcal{C}, J)}$ places the apex of the relative coend over U in the slice topos $\mathcal{E}/\ell_J(U)$. Pullback supplies the reindexing used in [equation \(4\)](#), so stable relative coends form cartesian objects of the canonical stack. This is the direct phase-1 link to relative topos theory. A further identification of such objects with a construction intrinsic to an arbitrary geometric morphism $\mathcal{F} \rightarrow \mathcal{E}$ requires a descent theorem and is not used here.

Drieux's indexed hom construction is the motivating special case. In the notation of the slides, its value above U has generalized components

$$[v : V \rightarrow U] \mapsto \mathbb{D}(V)(v^*Y, v^*X), \tag{7}$$

followed, when required, by sheafification. The 2026 slides then formulate an indexed colimit through universal properties on every localization and state an indexed co-Yoneda computation [[Dri26a](#), [Dri26b](#)]. Formula [equation \(7\)](#) supports the shape of [definition 3.1](#); the slides do not supply a published proof of a general relative-coend theorem, so none is inferred from them.

6.2 Relation to Shulman’s indexed profunctors

For an S -indexed symmetric monoidal category with indexed coproducts, Shulman defines a canceling tensor product. Under Beck–Chevalley and fibrewise coequalizer hypotheses, the composite of indexed profunctors is computed by a coend-shaped coequalizer [Shu13, Definition 2.1, Equation (3.8), and Lemma 3.2]. This is an established indexed coend calculus and must be treated as the principal comparison theory, not as an anticipated application of the present paper.

The two constructions have different inputs. Shulman’s formula begins with an indexed monoidal base and packages fibrewise coproducts together with change-of-base adjoints. Definition 3.1 instead begins with an indexed functor H and enlarges each cowedge over U to every object of $\mathcal{C}/U$. A substantive comparison theorem should identify hypotheses under which Shulman’s canceling tensor represents the localized cowedge functor. Until that theorem is proved, the two constructions are related models rather than interchangeable notation.

6.3 Presentation boundary

An internal category in $\mathcal{E}$, a stack on $(\mathcal{C}, J)$, and a pseudofunctor on a chosen site are connected by comparison results, but they are not definitionally identical. Likewise, equivalent sites can present the same topos without giving literally equal indexing categories. The definition in definition 3.1 is therefore a fixed-presentation construction. Its invariance under indexed equivalence (proposition 5.1) is a first step, but it is not yet a Morita or change-of-site invariance theorem.

7 Next stages and precise proof obligations

The following problems delimit the next stages. They are not theorems of this draft.

Problem 7.1 (Existence and computation). Give checkable hypotheses on $\mathbb{V}$ and $\mathbb{D}$ under which $\text{Cow}_U^{\text{rel}}(H; -)$ is representable for every U . The expected sufficient package has cocomplete fibres, cocontinuous reindexing, left adjoints $u_! \dashv u^*$, and Beck–Chevalley isomorphisms. Derive an explicit coequalizer or Grothendieck-construction formula, and prove that its canonical maps b_u are invertible.

Problem 7.2 (Comparison with indexed canceling products). When Shulman’s hypotheses hold, determine whether the localized relative coend is represented by his canceling tensor product. State separately the comparison on a fixed site and the additional descent needed for a stack of coefficients.

Problem 7.3 (Fubini). For a suitable indexed functor of four variables, formulate and prove the comparison between an iterated localized coend and a single coend over a product indexed category. Track both fibrewise interchange and Beck–Chevalley for the two layers of slice localization.

Problem 7.4 (Relative co-Yoneda). For the indexed hom object suggested by equation (7), identify the coefficient system and tensor needed to type the expression

$$\int_{\text{rel}}^A \mathbb{D}(A, X) \otimes F(A),$$

then prove a stable equivalence with $F(X)$. The proof must distinguish presheaf-level computation, sheafification, and descent.

Problem 7.5 (Relative profunctors). Use the existence and Fubini theorems to define composition of localization-sensitive indexed profunctors. Prove associativity and units, and compare the result with Shulman’s indexed profunctor equipment and Lee’s 2-categorical construction [Lee23].

Problem 7.6 (Site-presentation invariance). Let two small or small-generated sites present equivalent base topoi. Find the additional comparison data on $\mathbb{D}, \mathbb{V}, H$ for which their relative-coend objects correspond. A satisfactory theorem should be invariant under the relevant site Morita equivalence, not merely under a fibrewise equivalence chosen on one presentation.

Together, these obligations turn the phrase *coend calculus in indexed category theory* into a staged programme with a clear novelty boundary. The present paper supplies the universal object to be computed; the later stages must earn the calculus by proving existence, stability, interchange, and presentation independence.

References

- [CZ21] Olivia Caramello and Riccardo Zanfa. Relative topos theory via stacks. 2021.
- [Dri25a] Bruno Drioux. Indexed categories, fibrations, stacks and relative category theory. Lecture slides, Wuhan Autumn School on Topos and Applications, 24–28 September 2025, 2025.
- [Dri25b] Bruno Drioux. Indexed categories for relative topos theory. Lecture slides, LIX, 3 June 2025, 2025.
- [Dri26a] Bruno Drioux. Relative (co-)Yoneda lemma for categories indexed over a small-generated site. Lecture slides, 17 June 2026, 2026.
- [Dri26b] Bruno Drioux. Relative (co-)Yoneda lemma for categories indexed over a small-generated site. Lecture slides, joint work with Olivia Caramello, Topology, Algebra, and Categories in Logic, 28 July 2026, 2026.
- [Lee23] Seerp Roald Lee. Indexed profunctors over 2-categories, 2023.
- [Lor23] Fosco Loregian. Coend calculus, 2023. Version 7.
- [Shu13] Michael Shulman. Enriched indexed categories. *Theory and Applications of Categories*, 28(21):616–695, 2013.